

MATH 161, Autumn 2025
SCRIPT 4: The Topology of a Continuum

In this sheet we give a continuum C a topology. Roughly speaking, this is a way to describe how the points of C are ‘glued together’.

Definition 4.1. A subset of a continuum is *closed* if it contains all of its limit points.

Theorem 4.2. *The sets $\emptyset$ and C are closed.*

Proof of Theorem 4.2. We want to prove that $\emptyset$ and C contain all of their limit points.

Firstly, we want to prove that the $LP(\emptyset) = \emptyset$. $LP(\emptyset) = \emptyset$ as the emptyset contains no limit points. Therefore, we may write

$$LP(\emptyset) \subset \emptyset$$
$$\emptyset \subset \emptyset$$

which we know are equal.

Hence, we can say that $\emptyset$ contains all of its limit points and is therefore closed.

Secondly, we want to prove that the $LP(C) = C$. Suppose there is a limit point $p \in C$. As C has an ordering of $<$ and has no first and last point, all regions R that contain p have $R \cap (A \setminus \{p\}) \neq \emptyset$. Thus, as there exists some $p \in LP(C)$ for all $p \in C$, we may say that $C \subset LP(C)$. By the definition of limit points, we may say that $LP(C) \subset C$. Therefore, we may say that $LP(C) = C$. Hence, we can write

$$LP(C) \subset C$$
$$C \subset C$$

which we know are equal.

Hence, we can say that C contains all of its limit points and is therefore closed. □

Theorem 4.3. *A subset of C containing a finite number of points is closed.*

Proof of 4.3. We want to prove that a finite subset of C contains all of its limit points.

Let $A \subset C$ where A contains a finite number of points. A set with a finite number of points is a finite set, as one can draw a bijective correspondence with $[n]$ where n is the number of points in the set. By **Theorem 3.23**, we know that finite sets contain no limit points. Therefore, $LP(A) = \emptyset$ and $LP(A) \subset A$. As there are no limit points, we can say all limit points are contained by the set. This works to prove that a finite subset of C is therefore closed. □

Definition 4.4. Let X be a subset of C . The *closure* of X is the subset $\overline{X}$ of C defined by:

$$\overline{X} = X \cup LP(X).$$

Theorem 4.5. $X \subset C$ is closed if and only if $X = \overline{X}$.

Proof of Theorem 4.5. We want to prove that a subset of the continuum is closed if and only if the subset is equal to its closure.

Firstly, suppose that $X \subset C$ contains a finite set of points. By **Theorem 4.3**, we know that this set is closed. As X contains a finite set of points we may say that $LP(X) = \emptyset$. Therefore, $\overline{X} = X \cup LP(X) = X \cup \emptyset$. Therefore, if X contains a finite set of points, then we may say that $X \subset C$ is closed and $X = \overline{X}$.

Secondly, suppose that $X' \subset C$ is not closed. By **Definition 4.1** a set is closed if it contains all of its limit points $LP(X')$. Therefore the set $X' \cup LP(X')$ would contain all of its limit points. $X' \cup LP(X') = \overline{X'}$. Thus, we may say that only if $X' = X' \cup LP(X')$ then we may say it is closed and therefore, if and only if $X' = \overline{X'}$ we may say $X' \subset C$ is closed.

Hence, we have proven that $X \subset C$ is closed if and only if $X = \overline{X}$. □

Theorem 4.6. Let $X \subset C$. Then $\overline{X}$ is closed. (Equivalently, $\overline{X} = \overline{\overline{X}}$.)

Proof of Theorem 4.6. We want to prove that $\overline{X} = \overline{\overline{X}}$. We want to prove this by double inclusion.

Firstly, we want to prove that $\overline{X} \subset \overline{\overline{X}}$. By **Definition 4.4** we may write

$$\overline{X} \subset \overline{X} \cup LP(\overline{X})$$

$$\overline{X} \subset (X \cup LP(X)) \cup LP(\overline{X})$$

Now, we want to show that $LP(\overline{X}) = LP(X)$.

First, as $\overline{X} = X \cup LP(X)$ and $X \subseteq \overline{X}$, we may write $LP(X) \subseteq LP(\overline{X})$. Second, for the reverse inclusion, let $p \in LP(\overline{X})$. Then every region R containing p satisfies

$$R \cap (\overline{X} \setminus \{p\}) \neq \emptyset.$$

Since $\overline{X} = X \cup LP(X)$, we have Let $p \in LP(\overline{X})$. Every region R containing p meets

$$(\overline{X} \setminus \{p\}) = (X \setminus \{p\}) \cup (LP(X) \setminus \{p\}).$$

If $R \cap (X \setminus \{p\}) = \emptyset$, then because $R \cap (\overline{X} \setminus \{p\}) \neq \emptyset$, the point in this intersection must lie in $LP(X) \setminus \{p\}$. Call this point q . Then, $q \in LP(X) \setminus \{p\}$, but then q being a limit point of X forces R to meet $X \setminus \{p\}$, which creates a contradiction. Hence $p \in LP(X)$. Thus $LP(\overline{X}) \subseteq LP(X)$.

Therefore, as $LP(X) = LP(\overline{X})$ we may write $\overline{X} \subset (X \cup LP(X)) \cup LP(\overline{X})$ as

$$\overline{X} \subset (X \cup LP(X)) \cup LP(X)$$

$$\overline{X} \subset X \cup LP(X) \cup LP(X)$$

$$\overline{X} \subset X \cup LP(X)$$

Thus, this works to prove that $\overline{X} \subset \overline{\overline{X}}$.

Secondly, we want to prove that $\overline{\overline{X}} \subset \overline{X}$. We may write this as

$$\overline{X} \cup LP(\overline{X}) \subset \overline{X}$$

As we have already proven that $LP(\overline{X}) = LP(X)$ we may write that

$$(X \cup LP(X)) \cup LP(X) \subset \overline{X}$$

$$X \cup LP(X) \subset \overline{X}$$

By **Definition 4.4** we have that $\overline{X} = X \cup LP(X)$ and therefore we may say that $X \cup LP(X) \subset \overline{X}$ is true. Thus, we have proven the statement by double inclusion. $\square$

Definition 4.7. A subset G of a continuum C is *open* if its complement $C \setminus G$ is closed.

Theorem 4.8. *The sets $\emptyset$ and C are open.*

Proof of Theorem 4.8. We want to prove that C and $\emptyset$ are open sets.

Firstly, we prove that the $\emptyset$ is an open set by showing $C \setminus \emptyset$ is closed. $C \setminus \emptyset = C$ and by **Theorem 4.2** we know that C is a closed set. Therefore, we may say that the $\emptyset$ is an open set.

Secondly, we prove that C is an open set by showing $C \setminus C$ is closed. $C \setminus C = \emptyset$ and by **Theorem 4.2** we know that $\emptyset$ is a closed set. Therefore, we may say that the C is an open set. $\square$

The following is a very useful criterion to determine whether a set of points is open.

Theorem 4.9. *Let $G \subset C$. Then G is open if and only if for all $x \in G$, there exists a region R such that $x \in R \subset G$.*

Firstly, we want to prove that if G is open then it contains regions around the point. Suppose we have G which is open. We may say this set contains no first and last points as it does not contain all of its limit points and therefore there must be a element greater than and less than all elements in the set. Let $a, x, b \in G$. As there are 3 elements, there must be 1 element between the other 2, hence let $a < x < b$. We can create a region R such that $x \in R$. The region R includes the elements around x that includes a and b . Therefore, there exists a region such that $x \in R$ and as $x \in G$ and $R \subset G$ we may say that there exists a region such that $x \in R \subset G$. As there exists no first and last point, this occurs for all $x \in G$.

Proof of Theorem 4.9. □

Corollary 4.10. *Every region R is open. Every complement of a region, $C \setminus R$, is closed.*

Proof of Corollary 4.10. We want to prove that every region R is open and thereby every complement of a region R is closed.

Firstly, we want to show that $C \setminus R$ is closed. We want to prove this by contradiction. Let R be a region in C and $X = C \setminus R$. Suppose there exists a limit point $p \in X$. Let $x \in R$. By the definition of limit point, there must be a region U such that $x \in U \subset R$. However, as $X = C \setminus R$, $X \cap U = \emptyset$ and therefore $x \notin U$. This cannot occur as every region R containing limit point p must intersect $X \setminus \{x\}$. Therefore, we know that $x \notin R$ and therefore $x \in X$ or $x \in C \setminus R$. Therefore, we can say that $C \setminus R$ is closed.

Secondly, by the nature of $C \setminus R$ being closed, by **Definition 4.7** we know that R is open.

Hence, this completes the proof. □

Corollary 4.11. *Let $G \subset C$. Then G is open if and only if for all $x \in G$, there exists a subset $V \subset G$ such that $x \in V$ and V is open.*

Corollary 4.12. *Let $a \in C$. Then the sets $\{x \in C \mid x < a\}$ and $\{x \in C \mid a < x\}$ are open.*

Proof of 4.12. Let $a \in C$ and $X = \{x \in C \mid x < a\}$ and $Y = \{x \in C \mid a < x\}$. We want to prove that both X and Y are open sets by contradiction.

Suppose we have $X = \{x \in C \mid x < a\}$ which is not an open set. By **Corollary 4.10**, the complement $C \setminus X$ should not be closed. We may write the complement of X as $C \setminus X = \{x \in C \mid x \geq a\}$. This can be written as $C \setminus X = \{a\} \cup Y$. By **Theorem 4.5** we know that if the closure of the set equals set then it is closed. As Y contains all elements and all limit points apart from a , the closure of the set $\overline{Y} = LP(A) \cup Y = \{a\} \cup Y$.

Hence, as $C \setminus X = Y \cup \{a\} = \overline{Y}$. Therefore, we may say that the complement $C \setminus X$ is closed. This is a contrapositive to the statement that X is an open set as it requires $C \setminus X$ to not be closed.

We may analogously prove that Y is open through the same principle as the complement $C \setminus Y = \{a\} \cup X$. Therefore, this works to prove that the sets $\{x \in C \mid x < a\}$ and $\{x \in C \mid a < x\}$ are open. □

Theorem 4.13. *Let G be a nonempty open set. Then G is the union of a collection of regions.*

Proof of Theorem 4.13. We want to show that an open set is the union of a collection of region.

Let $G \subset C$ be a nonempty open set. By **Theorem 4.9** we know that if G is open then $\forall x \in G, \exists R$ such that $x \in R \subset G$. If there exists a region R_k for all $x \in G$, all regions should be a subset of G . Therefore we may say that

$$R_1, R_2, \dots, R_n \subset G$$

As $x \in R$, and $x \in G$ and $R \subset G$, we can write $x \in R \subset G$. By this, we can say as R is a subset of G and every $x \in R$ is also $x \in G$, the union of R will still be a subset of G .

$$R_1 \cup R_2 \cup \dots \cup R_n \subset G$$

This may be written as

$$\bigcup_{x \in G} R_x \subset G$$

As we know that $x \in R$ for all x , we may say that $\bigcup_{x \in G} x \subset \bigcup_{x \in G} R_x$. Furthermore, as it is for all $x \in G$ we may say that $\bigcup_{x \in G} x \subset \bigcup_{x \in G} R_x = G$ and thereby completing the proof. $\square$

Exercise 4.14. Do there exist subsets $X \subset C$ that are neither open nor closed?

Proof of 4.14. We want to prove the existence of a subset $X \subset C$ that is neither open nor closed. Let X be a subset of the continuum defined as $X = \{x \in C \mid x > 0 \text{ and } x \leq 1\}$. Furthermore, let $a, b \in C$ such that $b > a$. We may say that there exists a region $\underline{ab}$ such that $b > 0$ and $a < 0$. Thus, we can say that $0 \in \underline{ab}$ and $\underline{ab} \cap X \setminus \{0\} \neq \emptyset$.

Now we may say that $0 \in LP(X)$ and $0 \notin X$. Therefore, X does not contain all of its limit points and thus we may say it is not a closed set.

To prove that X is not an open set, we must show that $C \setminus X$ is not a closed set. $C \setminus X = \{x \in \mathbb{Q} \mid x \leq 0 \text{ and } x > 1\}$. Let $c, d \in C$ such that $d > c$. We may say that there exists a region $\underline{cd}$ such that $d > 1$ and $c < 0$. Thus, we can say that $1 \in \underline{cd}$ and $\underline{cd} \cap X \setminus \{0\} \neq \emptyset$.

Therefore, there exists a limit point $1 \in LP(C \setminus X)$, however, $1 \notin C \setminus X$. Thus, we may say that $C \setminus X$ is not a closed set as it does not contain the limit point 1. This means that X is not an open set.

Hence, we have proven that there exists $X \subset C$ that are neither open nor closed. $\square$

Corollary 4.15. Let $\{X_1, \dots, X_n\}$ be a finite nonempty collection of closed subsets of a continuum C . Then the union $X_1 \cup \dots \cup X_n$ is closed.

Proof of Corollary 4.18. Let $\{X_1, X_2, \dots, X_n\}$ be a finite nonempty collection of closed subsets of a continuum C . We want to show that the union of all sets is closed.

For any closed set X we know that $LP(X) \subset X$. Furthermore, we know that limit points are commutative over finite regions. Hence, we may write

$$LP\left(\bigcup_{i=1}^n X_i\right) = \bigcup_{i=1}^n LP(X_i)$$

By the definition of the closed set, we may now write

$$\bigcup_{i=1}^n LP(X_i) \subset \bigcup_{i=1}^n X_i$$

As the union of all limit points of all sets in the collection $\{X_1, X_2, \dots, X_n\}$ forms a subset of the collection, we may say that the union of all sets contains all of its limit points. We may write this as $LP(\bigcup_{i=1}^n X_i) \subset \bigcup_{i=1}^n X_i$. Hence, this proves that the union of all closed subsets of C is closed. $\square$

Exercise 4.16. Is it necessarily the case that the intersection of an infinite number of open sets is open? Is it possible to construct an infinite collection of open sets whose intersection is not open? Equivalently, is it possible to construct an infinite collection of closed sets whose union is not closed?

Proof for Intersections. Consider the collection of regions

$$\left(-\frac{1}{n}, \frac{1}{n} \right) \quad \text{for each } n \in \mathbb{N}.$$

Each $\left(-\frac{1}{n}, \frac{1}{n} \right)$ is open as all regions are open. However, the intersection of all these regions is

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n} \right) = \{0\}.$$

where the singleton set $\{0\}$ is not open, because any region containing 0 must contain points other than 0 to be open. Therefore, this proves that the intersection of infinitely many open sets need not be open. $\square$

Proof for Unions. For each $n \in \mathbb{N}$, the complement $C \setminus \left(-\frac{1}{n}, \frac{1}{n} \right)$ is closed as all regions are open. Consider the union of these complements:

$$\bigcup_{n=1}^{\infty} \left(C \setminus \left(-\frac{1}{n}, \frac{1}{n} \right) \right) = C \setminus \{0\}.$$

This union is not closed, since it does not contain the limit point 0. Therefore, an infinite union of closed sets need not be closed. $\square$

Theorem 4.13 says that every nonempty open set is the union of a collection of regions. This necessary condition for open sets is also sufficient:

Corollary 4.17. *Let $G \subset C$ be nonempty. Then G is open if and only if G is the union of a collection of regions.*

Proof of Corollary 4.20. We want to prove that G is open if and only if G is the union of a collection of regions.

Firstly, we want to show that if G is a nonempty, open set, then G is the union of a collection of regions. By **Theorem 4.13** we know this to be true.

Secondly, we want to show that if G is a union of a collection of regions, then G is a nonempty, open set. By **Corollary 4.10** we know that every region is open. Then, by **Corollary 4.16**, we know that the union of a set of open subsets of C is open. As we consider G to be the union of a collection of regions, we know that the set is nonempty, and we know that it is open.

This proves that if $G \subset C$ is nonempty, then G is open if and only if G is the union of a collection of regions. $\square$

Corollary 4.18. *If $\underline{ab}$ is a region in C , then $\text{ext } \underline{ab}$ is open.*

Proof of Corollary 4.21. Suppose there exists a region $\underline{ab}$ in C . We want to show that $\text{ext } \underline{ab}$ is open.

We define $\text{ext } \underline{ab}$ as

$$\text{ext } \underline{ab} = \{x \in C \mid x < a\} \cup \{x \in C \mid b < x\}$$

To show that $\text{ext } \underline{ab}$ is open, we want to show that each set in the union is open.

Firstly, we want to show that $\{x \in C \mid x < a\}$ is open. As $a \in C$, by **Corollary 4.12** we may say that this set is open.

Secondly, we want to show that $\{x \in C \mid b < x\}$ is open. Similarly, as $b \in C$ by **Corollary 4.12** we may say that this set is open.

Now, as the two sets are open, we may say that $\text{ext } \underline{ab}$ is the union of open sets and therefore is open. This works to prove that $\text{ext } \underline{ab}$ is open. $\square$

Definition 4.19. Let C be a continuum. We say that C is *disconnected* if it may be written as $C = A \cup B$, where A and B are disjoint, non-empty open sets. C is *connected* if it is not disconnected.

Theorem 4.20. *Let C be a connected continuum. Let $x, y \in C$, with $x < y$. Then there exists $z \in C$ such that $x < z < y$. In particular, all regions of a connected continuum are nonempty.*

Proof of Theorem 4.23. We want to prove that for any $x, y \in C$ where $x < y$, there exists some $z \in C$ such that $x < z < y$.

Suppose, for the sake of contradiction, that there exists no element $z \in C$ such that $x < z < y$. Now, define the sets

$$A = \{c \in C : c \leq x\} \quad \text{and} \quad B = \{c \in C : c \geq y\}.$$

Since $x < y$, both A and B are nonempty. Furthermore, by the construction of sets, we know that every element of C belongs to either A or B . Therefore, we may write C as $C = A \cup B$. Moreover, as A and B are disjoint, we may write them as $A \cap B = \emptyset$ and that every element of A is strictly less than every element of B .

This contradicts the pre-existing assumption that C is connected. Therefore, as we know the assumption is true, we may say that exists $z \in C$ such that $x < z < y$.

To prove the final statement, let $R \subset C$ be a region. By definition, there exist $x, y \in C$ with $x < y$ such that

$$R = \{c \in C : x < c < y\}.$$

By the first part of the proof, there exists $z \in C$ with $x < z < y$, and hence $z \in R$. Thus every region of a connected continuum is nonempty. $\square$

Exercise 4.21. Let C be a connected continuum and $a \in C$. Prove that $C \setminus \{a\}$ is a disconnected continuum.

Exercise 4.24. We want to prove that $C \setminus \{a\}$ is a disconnected continuum. As $C \setminus \{a\}$ is a continuum, we may assume that it is nonempty, has no first and last point, and has the same ordering as C .

Now, let $A = \{x \in C \mid x < a\}$ and $B = \{x \in C \mid x > a\}$ such that $C \setminus \{a\} = A \cup B$. Therefore, we may say that A and B are disjoint. By **Corollary 4.12** we know that A is open. Therefore, by **Corollary 4.20** we may say that there exists a region $R = \underline{cd}$ such that $x \in R \subset C$. There are two cases that may occur:

Case 1: $d \neq a$: Then $c < x < d < a$ and this completes the proof.

Case 2: $d = a$: Then, as we know there exists a point s such that $d < s < a$, we may create a region $\underline{cs}$ such that $c < x < s$. Therefore, there is a region that contains x in $C \setminus \{a\}$. Hence, A is open in $C \setminus A$.

By analogous reasoning for B , we have proven that $C \setminus \{a\}$ is a disconnected continuum. $\square$

Exercise 4.22. Must every realization of a continuum be disconnected? Think about the realizations of the continuum from Exercise 3.26. Are they connected/disconnected?

Additional Exercises

1. Prove that if $S \subset C$ then

$$\overline{S} = \{x \in C \mid \text{for all regions } R \text{ containing } x, R \cap S \neq \emptyset\}.$$

This is the best way to think of closure - in general, splitting into S and its limit points is a very inefficient approach.

2. Find an example of a continuum C and a subset that is both open and closed, other than $\emptyset$ and C .
3. Prove that if $X \subset Y$ then $\overline{X} \subset \overline{Y}$. If $\overline{X} = \overline{Y}$, is it necessarily true that $X = Y$?
4. Suppose $A, B \subset C$. Prove that

$$\overline{A \cap B} \subset \overline{A} \cap \overline{B}.$$

Can one expect equality in general? Why or why not?

5. Prove that the set of limit points of a set $A \subset C$ is a closed set.
6. Let $A \subset C$. We say that $x \in A$ is an *interior* point of A if there is a region R such that $x \in R \subset A$. We let $\text{int}(A) = \{a \in A \mid a \text{ is an interior point of } A\}$.
 - (a) Let $A = \underline{ab}$, for $a, b \in C, a < b$. Find $\text{int}(A)$.
 - (b) Show that A is open if, and only if, $A = \text{int}(A)$.
 - (c) Show that $\text{int}(A)$ is open.
7. Let A, B be subsets of C . Either prove or give a counterexample to each of the following:

$$\begin{aligned} \text{int}(A \cap B) &= \text{int}(A) \cap \text{int}(B) \\ \text{int}(A \cup B) &= \text{int}(A) \cup \text{int}(B). \end{aligned}$$

8. Let C be a continuum.
 - (a) Show that if A, B are subsets of C such that $A \subset B$ then, if B is closed, $\overline{A} \subset B$.
 - (b) Show that if A, B are subsets of C such that $A \subset B$ then, if A is open, $A \subset \text{int}(B)$.
 - (c) Let $A \subset C$ and $\mathcal{F}_A = \{B \subset C \mid B \text{ is a closed set containing } A\}$. Show that $\overline{A} = \bigcap_{B \in \mathcal{F}_A} B$.
 - (d) Formulate and prove a result analagous to c) for $\text{int}A$.
9. Prove that
 - i) $\{0\}$ is closed but not open in $\mathbb{Q}$.
 - ii) Prove that $\underline{01}$ is open but not closed in $\mathbb{Q}$.
10. Consider the set $X = \{x \in \mathbb{Q} \mid x \leq 0\} \cup \{x \in \mathbb{Q} \mid 0 < x \text{ and } x^2 \leq 2\}$. Determine whether this set is open/closed/both/neither in $\mathbb{Q}$.
11. Let $R = \underline{ab}$ be a region in C . Must it hold that $\overline{R} = \{x \in C \mid a \leq x \leq b\}$? Either prove or find a counterexample.
12. For each part, give an *explanation*, but you do not need to give a full proof.
 - (a) Give an example of a continuum C and a non-empty, closed set $A \subset C$ with no limit points.
 - (b) Give an example of a continuum C and a non-empty, closed set $A \subset C$ with exactly one limit point.

- (c) Give an example of a continuum C and a non-empty, closed set $A \subset C$ with a countably infinite set of limit points.
- (d) Give an example of a continuum C and a non-empty, open set $A \subset C$ with no limit points.
- (e) Give an example of a continuum C and a non-empty, open set $A \subset C$ with exactly one limit point.¹
- (f) Give an example of a continuum C and a non-empty, open set $A \subset C$ with a countably infinite set of limit points.

13. Let $X \subset C$. Then if $A \subset X$, we use $\overline{A}^X$ to denote the closure of A in X , i.e.

$$\overline{A}^X = A \cup \{x \in X \mid x \text{ is a limit point of } A\} = (\overline{A} \cap X).$$

We say that A is *closed in* X if $\{x \in X \mid x \text{ is a limit point of } A\} \subset A$. (Thus A is closed in X if it contains all its limit points in X .) We say that A is *open in* X if its complement is closed in X .

We define X to be disconnected if there are non-empty disjoint sets $A, B \subset X$ that are *open in* X such that $X = A \cup B$.

- (a) Prove that X is disconnected if, and only if, there are non-empty subsets A and B such that $X = A \cup B$ and $\overline{A}^X \cap B = \emptyset, \overline{B}^X \cap A = \emptyset$.
- (b) Prove that X is disconnected if, and only if, there are non-empty disjoint sets $A, B \subset X$ that are *closed in* X such that $X = A \cup B$.

¹Hint: There are no such examples if $C = \mathbb{Q}$ or $\mathbb{Z}$ (think about why). However, one may find an example by looking at a well-chosen subset of $\mathbb{Q}$.